

Conformal Repair on a Budget: An Empirical Protocol for Retraining Decisions

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Abstract

Distribution shift can compromise the empirical reliability of conformal prediction intervals, but a coverage failure does not by itself imply that the predictive model must be refit. This paper studies *budgeted conformal repair*, an empirical maintenance rule for finite samples that selects the first sufficient action under a prespecified intervention-cost ordering, subject to empirical coverage and interval-width constraints. Given an existing linear predictor, conformal intervals, labelled shifted repair data, a target coverage level, and a width budget, the rule compares no intervention, residual-scale recalibration, conformal recalibration, intercept-only updating, and a linear refit on update data. In a controlled Gaussian linear-regression testbed, the resulting minimal repair map separates no-failure cases, failures repaired by low-cost recalibration or bias updating, failures for which a full linear refit is the minimal sufficient action, and failures that remain unrepairable within the width budget. Additional mixed-shift experiments show that compounding residual inflation with bias or slope shift can move otherwise repairable failures into the unrepairable-within-budget category. Many reliability failures are repairable without refitting the predictor. High noise-scale shifts are often width-limited rather than refit-limited, while large pure slope shifts provide the clearest case where refitting becomes necessary. The contribution is a concrete empirical decision rule for distinguishing lower-intervention repair, refitting on update data, and no acceptable repair under a stated width budget.

Keywords: conformal prediction, distribution shift, recalibration, model maintenance

1. Introduction

Distribution shift can make predictive intervals empirically unreliable because the exchangeability conditions underlying standard conformal validity may fail (Barber et al., 2023). In deployed systems, distribution shift and performance degradation often raise the question of whether a model should be updated or refit (Ovadia et al., 2019; Regol et al., 2025). This paper asks a more specific maintenance question: *after conformal reliability degrades, which intervention is sufficient before paying the cost of the most expensive refit action?*

The answer depends on both reliability and efficiency. Conformal prediction provides marginal coverage under exchangeability (Vovk et al., 2005, Chapter 2, pp. 17–51), and split conformal calibration targets a chosen marginal coverage level (Angelopoulos and Bates, 2023, Section 1.1). Interval size is also an important evaluation criterion (Angelopoulos and Bates, 2023, Section 3.1). This paper therefore studies repair on a budget. An action is acceptable only if it restores empirical coverage while keeping the average interval width below a fixed inflation limit. Interval efficiency is thereby part of the repair decision, not a secondary diagnostic.

The setting is intentionally controlled. In a Gaussian linear-regression testbed, the paper isolates covariate shift, noise-scale shift, intercept shift, and slope shift, and then adds targeted mixed-shift experiments. For each shift and severity level, the rule reports one of four outcomes: no intervention is needed, a lower-cost interval or intercept repair is sufficient, a full model refit is the minimal sufficient action, or no candidate action satisfies both the coverage and width constraints. The distinction is that reliability failure, low-cost repairability, full-refit necessity, and unrepairability within budget are different events.

The paper makes three contributions:

- It formulates conformal model maintenance under shift as a budgeted minimal-repair problem, where an action must restore empirical coverage without exceeding a fixed interval-width budget.
- It introduces a cost-ordered minimal repair map, together with full-refit necessity, avoidable-refit, and unrepairability rates that keep infeasible repairs separate from evidence for refitting.
- It evaluates the decision rule across controlled synthetic shifts, mixed-shift experiments, repair-set and labelled-budget ablations, sensitivity analyses, and chronological real-data experiments.

The empirical protocol uses disjoint update, repair-calibration, selection, and test splits. Candidate repairs are fitted or calibrated before selection, and final performance is assessed only on held-out shifted data. The selected action is therefore an empirical maintenance decision. A deployment requiring a formal guarantee could select the action first and then recalibrate the selected fixed predictor on a fresh exchangeable calibration sample.

The protocol therefore focuses on the central question of the paper: *among the available repair actions, which one is the first sufficient action under the prespecified ordering without exceeding the width budget?*

The experiments test four concrete expectations. Covariate shift alone should often require no repair when $P(Y | X)$ is unchanged. Noise-scale shift should often be repairable by residual-scale or conformal recalibration, unless the required widening exceeds the width budget. Intercept shift should often be repairable by a low-dimensional intercept update rather than a full refit. Large pure slope shifts should provide the clearest cases where a full refit becomes necessary, because the conditional mean changes and interval-only repair can restore coverage only by excessive width inflation.

The remainder of the paper is structured as follows. Section 2 positions the work relative to conformal prediction under shift, recalibration, and model maintenance. Section 3 defines the initial conformal predictor and shifted data splits. Section 4 introduces the repair menu and the minimal sufficient intervention rule. Section 5 describes the synthetic and mixed-shift experiments. Section 6 gives a population-level Gaussian sanity check, and Section 7 reports the repair map, full-refit necessity rates, ablations, sensitivity analyses, and chronological real-data experiment. Section 8 discusses limitations and conformal interpretation.

2. Related Work and Positioning

Conformal intervals and exchangeability. Conformal prediction gives finite-sample coverage under exchangeability (Vovk et al., 2005; Shafer and Vovk, 2008; Angelopoulos and Bates, 2023). In regression, the absolute-residual construction used here fits a predictor on one sample and calibrates a residual quantile on another. The paper takes these intervals as the object to monitor and repair under shift.

Conformal prediction under shift. Standard conformal coverage guarantees rely on exchangeability between the calibration data and the future test point, an assumption that can fail under distribution shift. Existing work addresses this problem through covariate-shift corrections (Tibshirani et al., 2019), robust conformal validation under specified perturbations (Cauchois et al., 2024), adaptive and online conformal inference (Gibbs and Candès, 2021; Gibbs and Candès, 2024), conformal tests for detecting distributional change (Vovk et al., 2021), and broader analyses beyond exchangeability (Barber et al., 2023). Multi-environment predictive inference studies related questions about prediction across heterogeneous environments (Duchi et al., 2025). Our contribution is complementary in that we assume labelled shifted repair data are available and ask which maintenance action is sufficient under a stated width constraint.

Recalibration, adaptation, and model updating. Related work studies how predictive uncertainty can be recalibrated or adapted when the data-generating process changes (Romano et al., 2019; Guo et al., 2017; Ovadia et al., 2019). Model-maintenance work asks when deployed models should be refit under limited information and cost constraints (Regol et al., 2025). This paper connects these concerns by separating the detection of a reliability failure from the intervention that follows: interval-only repair, a low-dimensional predictor update, a full refit, or no acceptable repair under the width budget.

3. Setup and Data Splits

We begin with a baseline regression model that defines the unshifted conditional relationship used by the initial predictor:

$$Y = X^\top \beta + \varepsilon, \quad X \in \mathbb{R}^p, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2).$$

A baseline training sample $\mathcal{D}_{\text{train}}$ is used to fit an ordinary least-squares predictor $\hat{f}_0(x) = \hat{b}_0 + x^\top \hat{\beta}_0$. The conformal nonconformity scores are the absolute residuals on the independent calibration sample:

$$R_i = |Y_i - \hat{f}_0(X_i)|, \quad (X_i, Y_i) \in \mathcal{D}_{\text{cal}}.$$

These residual scores define the source-calibrated interval and its reference width, which later serve as the baseline for measuring interval inflation:

$$q_0 = R_{(k)}, \quad k = \lceil (1 - \alpha)(|\mathcal{D}_{\text{cal}}| + 1) \rceil, \quad \hat{C}_0(x) = [\hat{f}_0(x) - q_0, \hat{f}_0(x) + q_0], \quad W_0 = 2q_0.$$

If $k > |\mathcal{D}_{\text{cal}}|$, we set $q_0 = \infty$, giving an uninformative but conservative interval.

For compactness, write $c = (j, s)$ for one shift setting, where j denotes the shift type and s its severity level. Each setting has four independent shifted samples: $\mathcal{D}_{\text{update}}^c$ for point-predictor updates, $\mathcal{D}_{\text{repair}}^c$ for interval calibration after a candidate predictor is fixed, $\mathcal{D}_{\text{sel}}^c$

for choosing the least costly sufficient action, and $\mathcal{D}_{\text{test}}^c$ for final evaluation. For readability, we omit the superscript when only one setting is under discussion.

The baseline intervals are then evaluated empirically under shift. For a fixed candidate predictor, calibration on $\mathcal{D}_{\text{repair}}$ has the usual conformal interpretation under exchangeability with future test points. Once multiple actions are compared and selected, the selected action is treated as an empirical maintenance decision in the repair rule below.

4. Budgeted Repair Rule

Action menu. The repair rule compares five candidate actions. Each action produces a fixed predictor $\hat{f}_a$ and an interval rule $\hat{C}_a$:

- NO-OP leaves the baseline predictor $\hat{f}_0$ and baseline half-width q_0 unchanged.
- SCALE keeps $\hat{f}_0$ but rescales q_0 by the ratio between the mean absolute residual on the repair-calibration data and the corresponding mean on $\mathcal{D}_{\text{cal}}$. With means $\bar{R}_{\text{rep}}$ and $\bar{R}_0$,

$$q_{\text{scale}} = q_0 \frac{\bar{R}_{\text{rep}}}{\bar{R}_0}.$$

This is a scalar interval correction, not a new conformal validity guarantee.

- RECAL (conformal recalibration) keeps $\hat{f}_0$ and recomputes the conformal residual quantile on $\mathcal{D}_{\text{repair}}$.
- INTERCEPT recentres the baseline predictor by adding the average shifted residual estimated on $\mathcal{D}_{\text{update}}$:

$$\hat{f}_{\text{int}}(x) = \hat{f}_0(x) + \hat{\delta}, \quad \hat{\delta} = \text{mean}_{\mathcal{D}_{\text{update}}} \{Y - \hat{f}_0(X)\}.$$

The interval is then calibrated on $\mathcal{D}_{\text{repair}}$ after the point predictor is fixed.

- RETRAIN denotes refitting all linear coefficients in the same model class on $\mathcal{D}_{\text{update}}$, followed by conformal recalibration on $\mathcal{D}_{\text{repair}}$. It is the most intervention-heavy action in this paper’s menu, not a general claim about any deployment-specific model-refitting pipeline. The model is not refit on the union of the original training data and the update data.

For each setting c , the candidate actions are ordered from least to most intervention-heavy:

$$\text{NO-OP} \prec \text{SCALE} \prec \text{RECAL} \prec \text{INTERCEPT} \prec \text{RETRAIN}.$$

This order is a fixed intervention convention rather than a measured runtime or monetary cost. The order is specified before evaluation and is neither learned nor optimised from $\mathcal{D}_{\text{sel}}$. In an application, it should represent deployment-specific intervention costs or preferences. Since SCALE and RECAL both use labelled repair-calibration data, their local order encodes a preference for the simpler scalar update. The policy comparison in Section 7.4 reports the effect of changing the available actions and then reversing nearby actions, in this order.

Algorithm 1: Budgeted minimal-repair procedure

Input: Baseline predictor $\hat{f}_0$. Data splits $\mathcal{D}_{\text{cal}}, \mathcal{D}_{\text{update}}, \mathcal{D}_{\text{repair}}, \mathcal{D}_{\text{sel}}$. Thresholds α, ϵ, κ .

Output: The least costly sufficient action, or UNREPAIRABLE.

1. Construct candidate repairs. Construct

$$\mathcal{A} = (\text{NO-OP}, \text{SCALE}, \text{RECAL}, \text{INTERCEPT}, \text{RETRAIN})$$

in the stated cost order, using only the data permitted for each action.

2. Evaluate the candidates. For every $a \in \mathcal{A}$, compute $\text{Cov}_{\text{sel}}(a)$ and $\text{Infl}_{\text{sel}}(a)$ on $\mathcal{D}_{\text{sel}}$.

3. Identify sufficient actions. Define

$$\mathcal{S} = \{a \in \mathcal{A} : \text{Cov}_{\text{sel}}(a) \geq 1 - \alpha - \epsilon, \text{Infl}_{\text{sel}}(a) \leq \kappa\}.$$

4. Select the repair. if $\mathcal{S} \neq \emptyset$ then

return the first action in $\mathcal{S}$ under the stated cost order

else

return UNREPAIRABLE

end

Selection rule. For a fixed setting c and split $S \in \{\text{sel}, \text{test}\}$, write $\text{Cov}_S(a)$ for empirical coverage and $\text{Infl}_S(a)$ for average interval width divided by W_0 . An action is sufficient only if it restores empirical coverage while staying within the width budget:

$$\text{Cov}_{\text{sel}}(a) \geq 1 - \alpha - \epsilon \quad \text{and} \quad \text{Infl}_{\text{sel}}(a) \leq \kappa. \quad (1)$$

Among the sufficient actions, the selected repair is the first one in the cost order:

$$a_{\min}(c) = \min_{\prec} \{a : a \text{ satisfies (1)}\}. \quad (2)$$

Because sufficiency is assessed empirically on the finite selection split, $a_{\min}(c)$ is a finite-sample maintenance decision. Independent test-split summaries diagnose the selected action. A confidence-adjusted variant could replace the plug-in coverage threshold in (1) with a binomial lower confidence bound, but the present paper studies the simpler plug-in rule. If no action satisfies (1), then $a_{\min}(c) = \text{UNREPAIRABLE}$. Equation (2) separates low-cost repair, RETRAIN necessity, and unrepairability within the stated width budget.

A no-op failure occurs when NO-OP violates either the coverage or width constraint on $\mathcal{D}_{\text{sel}}$. RETRAIN is counted as necessary only when no-op fails and $a_{\min}(c) = \text{RETRAIN}$. If $a_{\min}(c) = \text{UNREPAIRABLE}$, the setting is unrepairable within the chosen budget and is not treated as evidence that RETRAIN is necessary. RMSE is reported only as a diagnostic of point-prediction quality. It is not part of the selection rule.

Let F denote the event that NO-OP fails on the selection split. Unless stated otherwise, RNR, ARR, and Unrep. are conditioned on F . RNR-repair is additionally conditioned on repairability. This keeps unrepairable trials separate from Retrain-necessary trials.

Algorithm 1 summarises the complete decision procedure. The candidate actions are defined above, and the held-out test split is used only to evaluate the selected action after selection.

Table 1: Synthetic shift mechanisms.

Shift	Mechanism	Severity grid	Expected pattern
Covariate	X mean/variance shift; $Y \mid X$ unchanged	$s \in \{0, 0.5, 1, 1.5, 2\}$	NO-OP
Noise-scale	Residual standard deviation multiplied by s	$s \in \{1, 1.25, 1.5, 2, 3\}$	SCALE or RECAL
Intercept	$Y = X^\top \beta + s\sigma + \varepsilon$	$s \in \{0, 0.5, 1, 1.5, 2\}$	INTERCEPT
Slope	$Y = X^\top (\beta + sv) + \varepsilon$	$s \in \{0, 0.25, 0.5, 1, 1.5\}$	RETRAIN at high severity

5. Synthetic Experiment

5.1. Synthetic mechanisms

The synthetic experiment uses $p = 10$, baseline noise $\sigma = 1$, and a baseline coefficient vector with nonzero entries $(1, 0.8, 0.6, 0.4, 0.2)$ in the first five coordinates and zeros elsewhere. The four shift mechanisms in Table 1 are controlled cases for the repair rule. The expected pattern is a hypothesis, not a forced outcome.

The baseline distribution and all non-covariate shifted distributions use $X \sim \mathcal{N}(0, I_p)$. Covariate shift changes the mean of X in the direction $d = \mathbf{1}_p / \sqrt{p}$ and increases diagonal variances linearly from 1 to $1 + 0.25s$, while keeping $Y \mid X$ unchanged. Noise-scale shift multiplies the residual standard deviation by s , and intercept shift adds the response offset $s\sigma$. Slope shift uses $\beta_s = \beta + sv$, where v is unit-normalised with entries proportional to $(0.2, 0.4, 0.6, 0.8, 1.0)$ in coordinates 6–10 and zeros elsewhere. This perturbs coordinates that are zero in the baseline coefficient vector, making the slope shift a clean conditional-mean change.

5.2. Mixed-shift experiment

The four pure mechanisms isolate one type of shift at a time. This is useful for interpreting the repair map, but deployment shifts rarely occur in isolation. We therefore add a targeted mixed-shift experiment that combines calibration-repairable and model-relevant changes without expanding the experiment into a full benchmark grid.

We consider two mixed settings. Let η_m denote the noise multiplier, δ_m the intercept shift, and γ_m the slope-shift magnitude. In both settings, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The first setting combines a constant bias shift with residual noise inflation, so that both recentring and interval widening may be relevant:

$$Y = X^\top \beta + \delta_m + \eta_m \varepsilon.$$

This setting tests whether a mixed but partly calibration-repairable shift can still be handled by low-cost repairs such as SCALE, RECAL, or INTERCEPT.

The second setting combines a conditional-mean slope change with residual noise inflation, so that both refitting and width inflation may compete:

$$Y = X^\top (\beta + \gamma_m v) + \eta_m \varepsilon.$$

This setting tests whether model misspecification combined with wider residual noise makes RETRAIN more frequently necessary, or instead makes the setting UNREPAIRABLE under the width budget.

Table 2: Mixed-shift severity values.

Severity m	0	1	2	3	4
δ_m/σ	0	0.5	1.0	1.5	2.0
γ_m	0	0.25	0.5	1.0	1.5
η_m	1.0	1.25	1.5	2.0	3.0

For both mixed settings, we use the same baseline model, data splits, target coverage, tolerance, width budget, action set, cost ordering, and Monte Carlo repetition count as in the synthetic experiment. The severity index $m \in \{0, 1, 2, 3, 4\}$ uses the paired values in Table 2.

The mixed-shift results are not intended as an exhaustive evaluation of all possible shift combinations.

5.3. Monte Carlo protocol and reproducibility

Each Monte Carlo repetition uses $n_{\text{train}} = 1000$, $n_{\text{cal}} = 1000$, $n_{\text{update}} = 500$, $n_{\text{repair}} = 500$, $n_{\text{sel}} = 1000$, and $n_{\text{test}} = 5000$. The update, repair-calibration, and selection splits represent the labelled shifted data available to the repair protocol. The larger test split is used only for held-out evaluation in the simulation, so that post-selection coverage, width inflation, and RMSE are estimated with smaller Monte Carlo error. It is never used to fit, recalibrate, or select an action. The target coverage is $1 - \alpha = 0.90$, with tolerance $\epsilon = 0.02$ and width budget $\kappa = 1.5$. We run 1000 repetitions per shift setting.

The simulation proceeds as follows: fit the baseline least-squares model on $\mathcal{D}_{\text{train}}$, compute q_0 on $\mathcal{D}_{\text{cal}}$, generate the shifted update, repair, selection, and test splits, construct all five candidate actions, select the least costly sufficient action on $\mathcal{D}_{\text{sel}}$, and report post-selection performance on $\mathcal{D}_{\text{test}}$. The random seeds and condition order are fixed so that each reported setting can be regenerated from the configuration files. Calibration and recalibration were implemented with *CREPES* (Boström, 2022), with a manual order-statistic check for the reported conformal quantiles.

All rates are computed over Monte Carlo repetitions, not over individual test observations. Coverage, width inflation, and RMSE are averaged on the held-out test split after the selected repair action has been chosen on the selection split. RMSE is retained only as a point-prediction diagnostic and is not part of the repair-selection rule.

The implementation records one row per Monte Carlo repetition, shift type, severity level, intervention, and evaluation split. The recorded quantities include empirical coverage, average interval width, width inflation, RMSE, coverage shortfall, sufficiency indicators, the selected minimal intervention, and post-selection test success. The reported tables are computed directly from these recorded outputs. Code, configuration files, random seeds, generated tables, and replicate-level outputs are provided in the accompanying reproducibility materials.

6. Population-Level Gaussian Sanity Check

This section gives a population-level calculation for the controlled Gaussian testbed. Its purpose is to explain the expected transitions in the empirical repair map, rather than to

provide a finite-sample guarantee for the data-dependent repair-selection rule. The repair map is driven by how each shift changes the absolute residual scores, which are the quantities calibrated by conformal prediction. For a centred Gaussian residual with standard deviation σ , the ideal symmetric 90% half-width is approximately 1.645σ . Thus, the width budget κ acts approximately as a bound on how much the relevant residual quantile may grow.

Under pure covariate shift with unchanged $P(Y | X)$, the residual law around a correctly specified baseline predictor is unchanged, so no structural interval repair is needed asymptotically. Under noise-scale shift, the residual standard deviation is multiplied by c , so the required half-width also scales by c . Repair is feasible within the width budget only if $c \leq \kappa$. This is why high noise-scale settings can be unrepairable rather than refit-necessary.

Under intercept shift $Y = X^\top \beta + \delta + \varepsilon$, the baseline predictor has residual $\delta + \varepsilon$. Interval-only recalibration can restore marginal coverage by increasing the residual quantile, but the required half-width grows with $|\delta|$. Intercept updating recentres the residuals and can therefore restore coverage with less interval inflation once the bias is large enough.

Under slope shift $Y = X^\top (\beta + \Delta) + \varepsilon$, assume that the baseline predictor is correct before the shift, with $X \sim \mathcal{N}(0, I_p)$ and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The residual then combines the original noise with an additional Gaussian error caused by the slope change, giving variance $\sigma^2 + \|\Delta\|^2$. Let $\lambda = \|\Delta\|/\sigma$, which measures the size of the slope change relative to the noise level. For symmetric Gaussian intervals, the width required without refitting increases by the factor $\sqrt{1 + \lambda^2}$. Interval-only repair therefore exceeds the width budget when $\lambda > \sqrt{\kappa^2 - 1}$.

Beyond this threshold, interval-only repair must either violate the width budget or leave coverage unrepaired, so refitting the slope can become the least costly sufficient action. This calculation is a population benchmark rather than a finite-sample guarantee for the empirical selection rule.

7. Results

We report results from 1000 Monte Carlo repetitions per synthetic shift setting. Candidate actions are constructed using only their allowed update and repair-calibration splits, selected on the selection split, and evaluated on the independent test split. The reported coverage, width inflation, and RMSE are therefore held-out post-selection diagnostics.

For proportions computed from Monte Carlo repetitions, we use binomial Monte Carlo standard errors. A single 1000-repetition setting has worst-case standard error at most 0.016, corresponding to a conservative 95% Monte Carlo interval of approximately ± 0.031 . Aggregate rates use larger denominators, except for quantities conditioned on no-op failures. Those conditional rates should be interpreted with their effective failure counts in mind.

7.1. Minimal Repair Map

In the minimal repair map in Table 3, each entry reports the modal selected action over Monte Carlo repetitions together with its selection frequency. The map should be read as a decision table: for a given shift structure and severity, it asks which action is least costly among those satisfying both empirical coverage and width constraints.

An entry labelled UNREPAIRABLE means that no candidate action, including RETRAIN, satisfied both constraints under the chosen width budget. The map separates the repair outcomes. Covariate shift is mostly handled by NO-OP, consistent with unchanged $P(Y | X)$

Table 3: Modal minimal repair by shift and severity.

Severity level	Covariate	Noise-scale	Intercept	Slope
0	NO-OP (0.943)	NO-OP (0.921)	NO-OP (0.949)	NO-OP (0.926)
1	NO-OP (0.936)	SCALE (0.906)	SCALE (0.817)	NO-OP (0.772)
2	NO-OP (0.941)	SCALE (0.413)	SCALE (0.746)	SCALE (0.786)
3	NO-OP (0.920)	UNREPAIRABLE (1.000)	INTERCEPT (0.913)	SCALE (0.819)
4	NO-OP (0.912)	UNREPAIRABLE (1.000)	INTERCEPT (0.892)	RETRAIN (0.893)

in the correctly specified linear model. Noise-scale shift is handled by scale repair at mild severities but becomes UNREPAIRABLE once the residual scale exceeds the width budget. Intercept shift transitions from interval repair to intercept updating, because widening intervals around a biased predictor eventually becomes inefficient. Slope shift is the only setting where RETRAIN becomes the modal minimal repair, and only at the highest severity. Thus, the map does not say that refitting is never useful. It says that RETRAIN is necessary only in the settings where lower-cost conformal or low-dimensional repairs fail the joint coverage-width criterion.

7.2. Refit Necessity and Unrepairability

Table 4 summarises the failure and repair rates, and its note fixes the abbreviations and denominators used in the later result tables.

Table 4: Failure, repair, and post-selection test performance.

Shift	Failure	RNR-all	RNR-repair	ARR	Unrep.	Test succ.	Cov.	Infl.	RMSE
Covariate	0.070	0.066	0.078	0.782	0.152	0.981	0.901	1.003	1.007
Noise-scale	0.816	0.008	0.020	0.395	0.596	0.965	0.900	1.208	1.219
Intercept	0.792	0.020	0.021	0.941	0.039	0.958	0.904	1.112	1.110
Slope	0.640	0.317	0.337	0.624	0.059	0.944	0.900	1.107	1.113

Note. Across result tables, Failure is the fraction of Monte Carlo repetitions, or real-data deployment settings, where NO-OP violates coverage or width on $\mathcal{D}_{\text{sel}}$. Repairable fail. is the fraction of all settings with no-op failure and at least one sufficient action. Conditional on no-op failure, RNR-all is the RETRAIN-necessity rate, denoted RNR in compact tables. RNR-repair conditions further on repairable no-op failures. ARR is the avoidable-refit rate, meaning repair by an action preceding RETRAIN, and Unrep. is the fraction with no sufficient candidate. Test succ. is the fraction of selected actions that satisfy both the coverage and width constraints on the held-out test split. Cov., Infl., and RMSE are held-out test diagnostics for settings with a selected action, excluding UNREPAIRABLE decisions. Cov. is coverage, Infl. is average width divided by baseline width, RMSE is a point-prediction diagnostic, N/A means that the relevant denominator is zero, and Map chg. counts modal entries changed relative to the baseline map.

The aggregate rates separate reliability failure from RETRAIN necessity. Noise-scale and intercept shifts fail often, but rarely require RETRAIN. Their RNR-all values are only 0.008 and 0.020, respectively. Intercept shift has the highest avoidable-refit rate, 0.941, showing that almost all no-op failures under this shift are repaired by lower-cost actions. Slope shift has the highest RETRAIN-necessity rate, with RNR-all = 0.317, but still has substantial avoidable-refit mass at lower severities. High noise-scale shift produces the largest unrepairable rate, 0.596, showing that some failures cannot be repaired within the width budget even by RETRAIN. These cases are not evidence that RETRAIN is necessary. They indicate that the chosen efficiency budget is too strict for the shifted noise level.

Table 5 reports the targeted mixed-shift experiment. Abbreviations and denominators follow Table 4. N/A denotes no repairable failures for the conditional quantities.

Table 5: Mixed-shift results.

Mixed shift	Severity	Failure	RNR-repair	ARR	Unrep.	Test succ.	Infl.	Modal action
Intercept + noise	0	0.074	0.033	0.784	0.189	0.990	1.003	NO-OP
Intercept + noise	1	1.000	0.009	0.967	0.024	0.968	1.352	SCALE
Intercept + noise	2	1.000	0.136	0.420	0.514	0.938	1.453	UNREPAIRABLE
Intercept + noise	3	1.000	N/A	0.000	1.000	N/A	N/A	UNREPAIRABLE
Intercept + noise	4	1.000	N/A	0.000	1.000	N/A	N/A	UNREPAIRABLE
Slope + noise	0	0.063	0.071	0.825	0.111	0.985	1.002	NO-OP
Slope + noise	1	1.000	0.012	0.944	0.045	0.973	1.278	SCALE
Slope + noise	2	1.000	0.575	0.161	0.621	0.908	1.471	UNREPAIRABLE
Slope + noise	3	1.000	N/A	0.000	1.000	N/A	N/A	UNREPAIRABLE
Slope + noise	4	1.000	N/A	0.000	1.000	N/A	N/A	UNREPAIRABLE

In this table, RNR-repair is conditional on repairable no-op failures, whereas ARR and UNREPAIRABLE are conditional on all no-op failures. Therefore, RNR-repair, ARR, and UNREPAIRABLE are not intended to sum to one.

The mixed-shift experiment shows that the repair map remains informative when clean mechanisms are partially entangled. At severity 1, both mixed settings are usually repairable by residual-scale recalibration, with test-success rates of 0.968 for intercept-noise shift and 0.973 for slope-noise shift. Moderate mixed failures therefore need not imply RETRAIN. At severities 3 and 4, both settings are fully UNREPAIRABLE because residual inflation exceeds the width budget.

The slope-noise setting is stricter. At severity 2, RETRAIN is often selected among repairable failures, with RNR-repair = 0.575, but the unrepairable rate is also high at 0.621. The same severity setting can therefore show both frequent RETRAIN selection among feasible repairs and a substantial proportion of failures that cannot be repaired within the prescribed efficiency budget.

7.3. Sensitivity to the Coverage and Width Budgets

Because the decision rule depends directly on ϵ and κ , Table 6 reports a sensitivity analysis over $\epsilon \in \{0.01, 0.02\}$ and $\kappa \in \{1.25, 1.5, 2.0\}$. The no-op failure rate is 0.660 for $\epsilon = 0.01$ and 0.579 for $\epsilon = 0.02$ across this grid, so the table focuses on how failed cases are classified under different width budgets. The qualitative conclusions are stable: high noise-scale severity remains unrepairable at level 4, stricter budgets increase unrepairability, and relaxing the width budget reduces the frequency with which high slope shift requires RETRAIN. Abbreviations and denominators follow Table 4.

Table 6: Sensitivity to coverage tolerance and width budget.

ϵ	κ	RNR-all	ARR	Unrep.	Test succ.	Map chg.	Slope L4 retrain
0.01	1.25	0.146	0.454	0.400	0.903	4	0.749
0.02	1.25	0.172	0.486	0.342	0.959	3	0.893
0.01	1.50	0.095	0.597	0.308	0.907	1	0.749
0.02	1.50	0.099	0.657	0.244	0.962	0	0.893
0.01	2.00	0.036	0.759	0.205	0.911	2	0.091
0.02	2.00	0.018	0.838	0.144	0.964	3	0.050

The sensitivity analysis clarifies the role of the width budget. With $\kappa = 1.25$, more failures are classified as unrepairable because even successful coverage repair is too wide. With $\kappa = 2.0$, avoidable refits increase and high slope severity is less often classified as refit-necessary. These changes are expected rather than problematic: the paper studies RETRAIN necessity relative to a fixed efficiency budget.

7.4. Alternative maintenance policies and labelled-budget ablations

The preceding sensitivity analysis varies the numerical coverage and width budgets. We next compare alternative maintenance policies represented by different action sets and cost orderings. Using the same candidate-action metrics as in the synthetic experiment, we recompute the least costly sufficient repair after removing actions or changing the order of nearby actions. This isolates the effect of the maintenance policy without rerunning the data-generating process. We then separately rerun the full protocol with smaller labelled update, repair-calibration, and selection splits. Table 7 reports the policy comparison, and Table 8 reports the labelled-budget analysis. Abbreviations and denominators follow Table 4.

Table 7: Effect of changing the repair set.

Variant	RNR-all	RNR-repair	ARR	Unrep.	Test succ.	Map chg.
Full set	0.099	0.131	0.657	0.244	0.962	0
No Scale	0.115	0.155	0.624	0.261	0.959	6
No Recal	0.103	0.138	0.648	0.249	0.962	1
No Intercept	0.258	0.346	0.488	0.254	0.963	2
No Retrain	0.000	0.000	0.657	0.343	0.963	1
Interval only	0.000	0.000	0.488	0.512	0.963	4
Recal before Scale	0.099	0.131	0.657	0.244	0.960	6
Retrain before Intercept	0.258	0.341	0.488	0.244	0.962	2

The action-set ablation shows that the conclusions are not driven by a single recalibration option, but that they depend strongly on whether low-dimensional model repair is available. Removing residual-scale recalibration modestly increases RNR-all from 0.099 to 0.115 and the unrepairable rate from 0.244 to 0.261. Removing conformal recalibration has little effect on aggregate RETRAIN necessity. In contrast, removing the intercept-only update increases RNR-all from 0.099 to 0.258 and reduces the avoidable-refit rate from 0.657 to 0.488. The modal-map changes reveal that the high-severity intercept-shift settings that are repaired by INTERCEPT under the full set are promoted to RETRAIN when the intercept action is unavailable.

The *No Retrain* and *Interval only* rows should be read as availability checks rather than as evidence that refitting is unnecessary. Their RNR values are zero by construction because RETRAIN is removed from the set. The informative quantities are the increases in unrepairability from 0.244 under the full set to 0.343 without RETRAIN, and to 0.512 when only no-op, scale recalibration, and conformal recalibration are available. Thus, conformal recalibration alone repairs many failures, but not all failures that are repairable by lightweight model updating or RETRAIN. The ablation therefore separates failures repaired by recalibration, failures requiring model refitting, and failures that no available action can repair within the width constraint.

Finally, reversing the order of RETRAIN and intercept updating reproduces the No-Intercept pattern. RNR-all increases to 0.258. This confirms that cost ordering is not cosmetic. If RETRAIN is allowed to precede a lower-cost low-dimensional update, the rule can manufacture apparent RETRAIN necessity. Overall, the ablation supports the operational interpretation of the method: RETRAIN necessity is defined relative to the available action set, cost ordering, and width budget, not by the shifted distribution alone.

We also vary the amount of labelled shifted data available for repair. This ablation reruns the full protocol with smaller update, repair-calibration, and selection splits while keeping the held-out test split fixed. It therefore tests the finite-labelled-budget part of the proposed maintenance protocol, rather than merely recomputing decisions from the same candidate-action metrics. Table 8 reports the budget ablation. Abbreviations and denominators follow Table 4.

Table 8: Effect of labelled repair budget.

Budget	Failure	RNR-all	RNR-repair	ARR	Unrep.	Test succ.	Cov.	Infl.
100/100/200	0.603	0.125	0.184	0.554	0.321	0.915	0.903	1.095
250/250/500	0.587	0.108	0.149	0.612	0.281	0.945	0.902	1.093
500/500/1000	0.579	0.099	0.131	0.657	0.244	0.962	0.901	1.093

The labelled budget ablation shows that reducing the available shifted labelled data makes repair selection less stable. Moving from the full budget 500/500/1000 to 100/100/200 increases the no-op failure rate from 0.579 to 0.603, increases the unrepairable rate from 0.244 to 0.321, and reduces held-out test success from 0.962 to 0.915. Because the shifted distributions are unchanged, the change in no-op failure reflects noisier empirical classification on smaller selection splits rather than a different deployment problem.

RNR-all also rises from 0.099 to 0.125, while ARR falls from 0.657 to 0.554. Thus, smaller labelled repair budgets make repair decisions more conservative and less reliable on the held-out test split. These results show that RETRAIN necessity is indexed not only by the action set, cost ordering, and width budget, but also by the amount of labelled shifted data available for update, recalibration, and selection.

7.5. Real-data chronological experiment

The synthetic experiment is the controlled test because the shift mechanism is known by construction. We also evaluate the same decision rule on chronological forecasting data, where the shift mechanism is unknown. This real-data experiment is a compact diagnostic of the repair rule under temporal distribution change, not a benchmark study or an attempt to identify the underlying shift type.

We use three time-series forecasting data sets from the Monash forecasting archive (Godaheva et al., 2021): Electricity, San Francisco Traffic, and Solar Energy, available through forecastingdata.org. Solar is aggregated from 10-minute resolution to hourly resolution by averaging the observations within each hour. Evaluation is restricted to daylight target hours to avoid trivial night-time zero-output periods. Each univariate series is converted into a supervised lag-regression problem. Let τ denote the forecast origin and set the forecast horizon to $H = 24$. For series i , the supervised response is $Y_{i,\tau} = y_{i,\tau+H}$. The feature vector contains the available lags $y_{i,\tau-\ell}$ for $\ell \in \{1, 2, 24, 48, 168\}$, together with sine-cosine

encodings of hour-of-day and hour-of-week for the target time $\tau + H$. Rows with unavailable lag values, unavailable target values, or non-finite values are discarded before fitting and evaluation. Thus the real-data experiment remains a linear lag-regression forecasting problem, not a specialised time-series model. Each series is standardised using the mean and standard deviation computed from its initial training portion only. The same transformation is then applied to the initial calibration and later deployment examples before examples are pooled across sampled series.

For each repetition, we sample up to 80 usable series from the data set. A series is considered usable if it contains enough observations after preprocessing to form the initial training, initial calibration, deployment-window, lag, and horizon splits. The first 35% of each series is used for initial training and the next 15% for initial conformal calibration. The remaining 50% is treated as a later deployment period and divided into five relative chronological deployment windows. Within each deployment window, the first 20% is used for point-predictor updates, the next 20% for repair calibration, the next 20% for selecting the least costly sufficient action, and the final 40% for held-out evaluation. Thus, with 250 repetitions and five deployment windows, each data set contributes 1250 deployment settings. Repetitions differ through the sampled subset of usable series. All chronological split fractions and repair-selection rules are fixed across repetitions.

The point predictor is a ridge-stabilised linear model fit on the initial training data. The ridge penalty was fixed at 1.0, with the intercept unpenalised. The same penalty was used for the initial fit and all RETRAIN refits solely to stabilise the linear fit. It was not part of the repair-selection rule. We then apply the same five cost-ordered actions as in Section 4. Scale recalibration rescales the baseline conformal half-width by the ratio of mean absolute residuals on the repair-calibration data and the initial calibration data. Conformal recalibration recomputes the conformal interval on the repair-calibration data while keeping the baseline point predictor fixed. Intercept updating estimates only a global bias correction on the update data and then recalibrates on the repair-calibration data. The RETRAIN action is the same ridge-stabilised linear refit on the update data followed by recalibration on the repair-calibration data.

Actions are selected on the selection data using the same sufficiency criterion as in Equation (1), with $\alpha = 0.1$, $\epsilon = 0.02$, and $\kappa = 1.5$. Table 9 reports the real-data decision summary for this width budget. Final coverage, width inflation, and RMSE are evaluated on the held-out test part of each deployment window, but the compact table focuses on the repair decision categories. Abbreviations and denominators follow Table 4. In Table 9, Failure and Repairable fail. are fractions of deployment settings, while RNR, ARR, and UNREPAIRABLE are conditional on no-op failure. The modal repair action is computed among repairable no-op failures.

Table 9: Chronological real-data experiment.

Data set	Failure	Repairable fail.	Modal repair action	RNR	ARR	UNREPAIRABLE
Electricity	0.380	0.090	SCALE	0.015	0.223	0.762
Traffic	0.225	0.073	SCALE	0.028	0.295	0.676
Solar	1.000	0.970	RECAL	0.006	0.963	0.030

Electricity and Traffic contain many no-op failures that are unrepairable under the strict width budget, while the repairable failures are usually handled by scale recalibration. Solar is the contrasting case: nearly all chronological settings fail under no-op, but most failures are repairable by conformal recalibration rather than RETRAIN. Across all three data sets, the RETRAIN-necessity rate remains small relative to the avoidable-refit and unrepairable categories. High unrepairability is a width-budget finding, not a failure of the repair rule.

8. Discussion and Limitations

The results point to the distinction that empirical reliability failure is not the same as RETRAIN necessity. Some failures are repaired by recalibration, some by a low-dimensional intercept update, some by RETRAIN, and some are infeasible under the stated width budget. The width constraint is central. Without it, coverage can often be restored by widening intervals, but the resulting intervals may be operationally unusable.

The interpretation is conditional on the repair rule. Intercept updating becomes minimal only when interval-only repair around a biased predictor is inefficient. The RETRAIN refit becomes minimal mainly when the conditional mean changes enough that earlier actions in the prespecified ordering fail the joint coverage-width criterion. High noise-scale failures have the different interpretation that if the irreducible residual scale exceeds the width budget, this refit cannot make intervals both reliable and narrow, so the appropriate label is UNREPAIRABLE for that budget.

The conformal interpretation is deliberately empirical under shift. Calibrated intervals are not treated as distribution-free valid after the deployment distribution changes. For a fixed candidate predictor, recalibration on exchangeable repair-calibration data has the usual conformal interpretation for future observations from the same distribution. Since the action is selected on a separate selection split, the reported test results are held-out post-selection diagnostics. A deployment requiring a formal guarantee for the final selected action could select the action first and then recalibrate the selected fixed predictor on an additional fresh exchangeable calibration sample. The width-budget constraint would then need to be checked again on the recalibrated interval. A more conservative empirical variant could also use confidence-adjusted coverage thresholds during selection.

The synthetic experiment is linear and controlled by design, which makes the repair mechanisms interpretable but limits claims about broader deployment settings. The chronological forecasting experiment is therefore included as a compact evaluation of the decision rule, not as evidence about a particular real-world shift mechanism. Extensions to non-linear predictors, heteroscedastic noise, dependent data, delayed labels, and refitting on the union of original training and update data may change the minimal repair map.

9. Conclusion

A conformal coverage failure does not by itself say what should be repaired. Failure may come from interval scale, predictor bias, conditional-mean change, or a too-strict width budget. This paper turns that ambiguity into a repair-selection problem.

In the controlled experiments, covariate shift was usually handled by no action, mild noise-scale and intercept shifts by recalibration or intercept updating, and large slope shifts

gave the clearest case for RETRAIN. Severe noise-scale shifts were often UNREPAIRABLE under the chosen budget, showing that some failures are efficiency-limited rather than model-refit problems.

The contribution is an empirical maintenance rule for conformal prediction under shift. Given an action set, cost order, labelled repair budget, coverage tolerance, and width budget, the rule reports the least costly sufficient action and separates RETRAIN-necessary cases from avoidable repairs and unrepairable failures. The takeaway is that conformal model maintenance is better framed as a budgeted repair problem than as a binary choice between accepting degraded coverage and refitting the model.

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